Aryan Rajendra Suthar

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SUMMARY:

Software Engineer with 3+ years of experience building scalable backend and distributed systems using Java, Python, Node.js, and cloud-native microservices architecture. Designed high-performance REST/gRPC APIs, event-driven systems with Kafka, and containerized platforms using Docker, Kubernetes, and AWS supporting 10K+ daily requests and 5,000+ business transactions. Strong expertise in system design, cloud infrastructure, database optimization, caching strategies, and CI/CD automation for large-scale production systems.

EDUCATION:

Master of Science in Software Engineering | Arizona State University, USA **| Grade: 3.50/4.00** **Aug 2024 - May 2026**
Bachelor of Engineering in Computer Engineering | S.K.N.C.O.E., India | **Grade: 3.00/4.00** **Aug 2019 - May 2023**

SKILLS:

Programming Languages: Java, Python, C++, C#, SQL, JavaScript, TypeScript, Bash
Frameworks & Backend: Spring Boot, FastAPI, Django, Node.js, Express.js, .NET Core
Frontend: React, Next.js, TypeScript, HTML5, CSS3
Cloud & Distributed Systems: AWS (Lambda, EC2, S3, SNS, SES, IoT Core), Azure (Functions, AKS, Blob Storage, Service Bus), Microservices Architecture, Serverless Computing, Distributed Systems
Streaming & Messaging: Apache Kafka, Event-Driven Architecture, Asynchronous Messaging
Databases & Caching: MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, Query Optimization, Indexing
Machine Learning & AI: Machine Learning, Deep Learning, LLMs, RAG, PyTorch, TensorFlow, MLflow, NLP, Computer Vision
Security: OAuth2, JWT, RBAC Authorization, TLS Encryption
DevOps & Infrastructure: Docker, Kubernetes, CI/CD Pipelines, Git, Azure DevOps, Linux
Software Engineering Concepts: System Design, REST APIs, gRPC, Object-Oriented Design, Scalable Systems, Observability, Logging, Monitoring, Unit Testing, Integration Testing, Performance Optimization

PROFESSIONAL EXPERIENCE:

Jayraj Fabrication, India | Software Engineer **May 2023 – May 2024**

- Architected a scalable cloud-native ERP platform using microservices architecture, design patterns, and distributed system principles with Python, Java (Spring Boot), and Node.js, processing 5,000+ enterprise transactions.
- Designed high-performance REST APIs and gRPC services using optimized data structures and algorithmic query optimization integrated with MySQL and PostgreSQL, serving 10,000+ daily requests and reducing turnaround time from 48 hours to under 6 hours.
- Implemented event-driven architecture using Apache Kafka with asynchronous messaging, fault tolerance mechanisms, and distributed tracing pipelines for resilient microservice communication.
- Optimized database indexing, caching strategies, and query algorithms, reducing backend latency 40% and improving large-scale API performance.
- Built Docker containerized services with CI/CD pipelines, enabling scalable infrastructure deployments, load balancing readiness, and automated production releases.
- Implemented secure authentication architecture using OAuth2, JWT, RBAC, and TLS encryption, strengthening enterprise-grade distributed system security.
- Integrated observability pipelines including distributed tracing, structured logging, and monitoring dashboards, improving debugging efficiency and incident response.
- Designed fault-tolerant distributed backend services with horizontal scalability, load balancing, and high-availability architecture across production environments.

Ignite Infosys, India | Software Engineer **Jan 2021 – May 2023**

- Developed Spring Boot and .NET Core microservices deployed on AWS and Azure, supporting distributed enterprise applications with 100K+ active users using scalable system architecture and design patterns.
- Designed scalable RESTful APIs using optimized algorithms, efficient data structures, and caching strategies, improving backend throughput and reducing latency.
- Built Apache Kafka streaming pipelines processing 50K+ events daily, enabling reliable asynchronous messaging and fault-tolerant distributed communication.
- Containerized services using Docker and Kubernetes, enabling horizontal scalability, load balancing, and high availability deployments.
- Developed React + TypeScript frontend modules, improving usability and enabling responsive distributed application interfaces.
- Implemented CI/CD pipelines and automated testing pipelines, improving deployment reliability and engineering productivity.
- Designed distributed tracing pipelines, centralized logging systems, and monitoring dashboards, improving system observability and production debugging.
- Collaborated in Agile engineering teams delivering 5+ backend services, participating in system design discussions, code reviews, and architectural decisions.

PROJECTS:

Physio – AI Powered Rehabilitation Platform | Node.js, FastAPI, React, Next.js, Redis, MySQL, AWS, Docker

- Designed a cloud-native healthcare platform using microservices architecture, scalable system design, and distributed computing principles for physiotherapy recommendation systems.
- Developed backend APIs using Node.js and FastAPI, supporting real-time generation of personalized rehabilitation programs.
- Built React + Next.js dashboards using TypeScript, enabling dynamic exercise visualization, analytics, and therapy management.
- Implemented Redis distributed caching strategies and algorithm optimization, reducing API latency 40% under concurrent workloads.
- Designed normalized relational database schemas and optimized data structures supporting scalable patient data management.

Distributed AI Simulation Platform | Python, Microservices, Kafka, Redis, Docker, Kubernetes, gRPC

- Architected a distributed simulation platform using microservices, event-driven architecture, and scalable system design principles, supporting 100+ concurrent simulation workloads.
- Implemented gRPC communication optimized with efficient serialization algorithms, improving inter-service efficiency 35% vs REST APIs.
- Built Kafka streaming pipelines for asynchronous event processing across distributed services.
- Implemented Redis caching strategies reducing redundant computation and improving throughput by 30%.
- Containerized services using Docker and Kubernetes, enabling scalable deployments and high availability.

Emofusion – Multimodal Emotion Detection System | Python, TensorFlow, PyTorch, MLflow, NLP, Computer Vision

- Developed a deep learning pipeline using optimized feature extraction algorithms and scalable ML data processing pipelines.
- Processed 61,000+ training samples through optimized feature extraction pipelines.
- Achieved 82% emotion classification accuracy using PyTorch neural network models.
- Implemented MLflow experiment tracking and model versioning for reproducible ML workflows.

AI Knowledge Assistant | Python, LangChain, OpenAI API, FastAPI, Redis, Vector Database

- Built a Retrieval-Augmented Generation system using vector databases, semantic search algorithms, and distributed inference pipelines.
- Implemented vector embeddings and semantic search pipelines for efficient document retrieval.
- Developed FastAPI backend integrating LLM inference pipelines using OpenAI APIs.
- Optimized LLM inference pipelines and caching strategies, reducing latency 30% (<100ms).
- Implemented Redis caching and session management enabling scalable AI inference workloads.